

Data Ingress & Egress

Where data enters and leaves each part
of the Linear pipeline (today vs target)

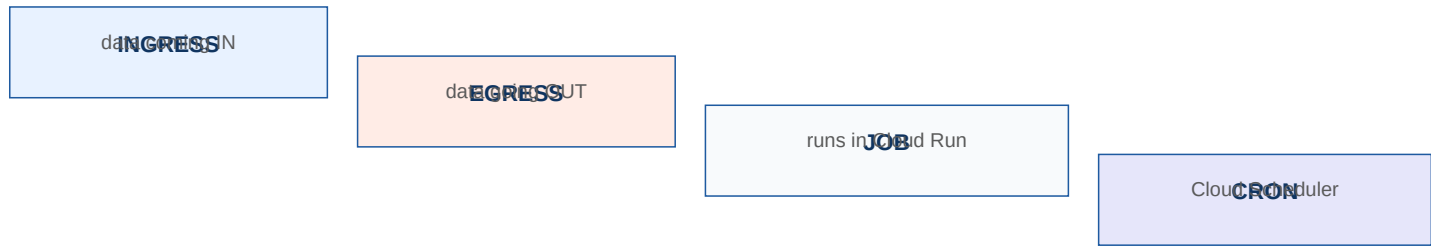
GCP project:	peq-tools
Focus:	What the cron jobs actually touch
Date:	June 2026

Contents

- 1 What ingress and egress mean
- 2 Quick answer: what does cron do today?
- 3 TODAY: 9 AM email job (live)
- 4 TODAY: Copy job (built, manual only)
- 5 TODAY: BigQuery (not connected yet)
- 6 TARGET: full pipeline with two crons
- 7 Side-by-side comparison table
- 8 Where to study in the repo

1. What ingress and egress mean

Every piece of the pipeline has data flowing IN and data flowing OUT. Leaders often ask for this map so they can see what each job actually touches.



Simple definitions

- Ingress = data entering a component (reads, downloads, fetches)
- Egress = data leaving a component (writes, uploads, sends)
- A job sits in the middle: it pulls data in, transforms it, pushes data out

Key idea

The pipeline diagram shows boxes. This document shows the ARROWS - what crosses each boundary and over which channel (API, email, files, database).

2. Quick answer: what does cron do today?

Today: only ONE cron job

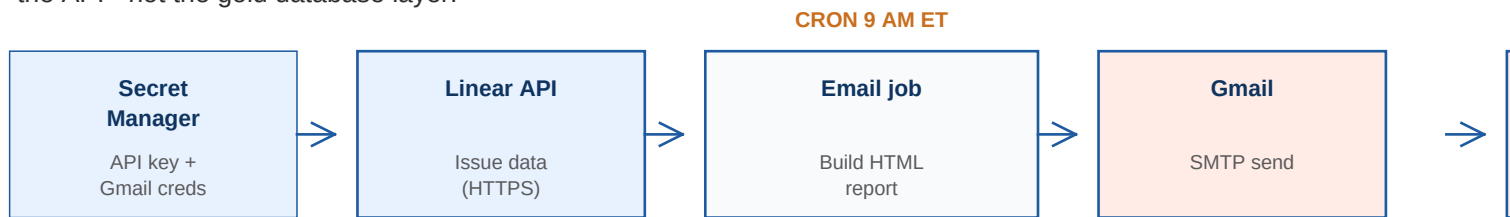
Cloud Scheduler runs daily-sweep-report at 9:00 AM Eastern, Mon-Fri. It sends the sweep email. It does NOT run the Linear data copy.

Scheduled vs manual today

Component	Trigger	Ingress (in)	Egress (out)
daily-sweep-report	CRON 9 AM	Secret Mgr + Linear API	Gmail to inbox
linear-ingest (copy)	Manual only	Secret Mgr + Linear API	GCS bronze files
BigQuery silver/gold	N/A	Nothing yet	Nothing yet

3. TODAY: 9 AM email job (live)

Job name: daily-sweep-report. Runs on Cloud Run in GCP project peq-tools. Currently uses a test Linear workspace via the API - not the gold database layer.



Ingress detail (what comes IN)

- peq-tools-daily-sweep-report-config: Linear API key, Gmail address/password, recipients
- Linear GraphQL API: issues matching yesterday's sweep window

Egress detail (what goes OUT)

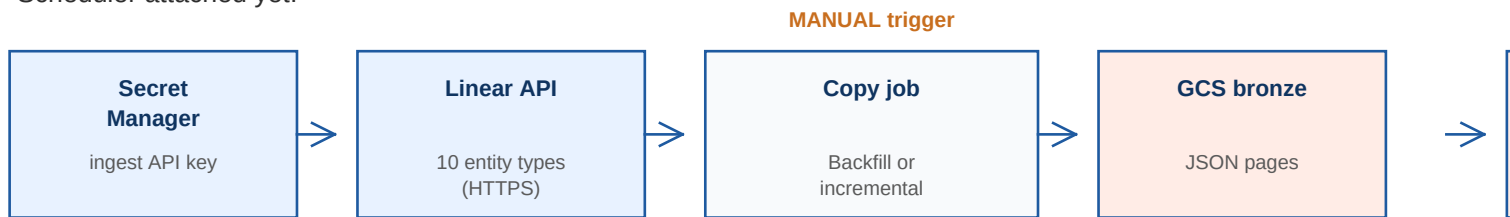
- One HTML email per run via Gmail SMTP (ports 465/587 from cloud - works)

What this job does NOT touch today

- GCS bucket peq-tools-linear-data (bronze archive)
- BigQuery linear_silver or linear_gold
- peq-tools-linear-ingest-config secret

4. TODAY: Copy job (built, manual only)

Job name: linear-ingest. Same Cloud Run platform, different command. Validated locally and in cloud - but NO Cloud Scheduler attached yet.



Ingress detail

- peq-tools-linear-ingest-config: Linear API key (read-only)
- Linear API: organization, teams, issues, users, comments, etc.

Egress detail

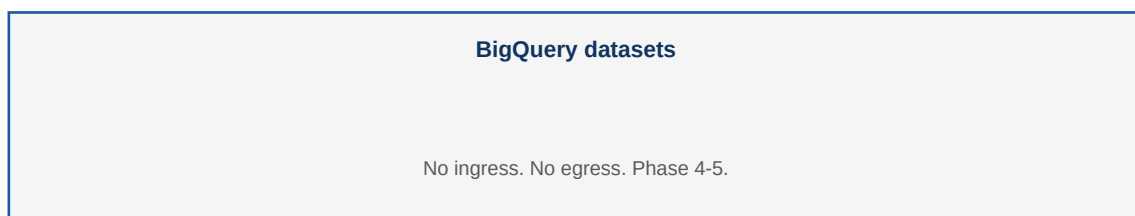
- gs://peq-tools-linear-data/bronze/backfill|daily/run_.../page_*.json
- gs://peq-tools-linear-data/state/watermarks.json (updated each successful run)
- Per-run summary.json with record counts and status

Incremental / watermark behavior

- backfill = full history once; incremental = only records changed since bookmark
- Reference entities (teams, labels) still full-snapshot each run

5. TODAY: BigQuery (not connected yet)

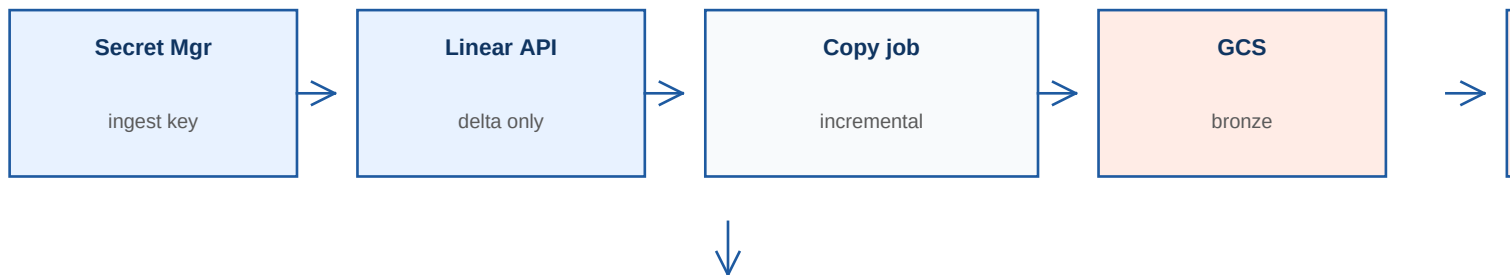
Terraform created datasets linear_bronze, linear_silver, linear_gold. They are empty shells. No job reads or writes them yet.



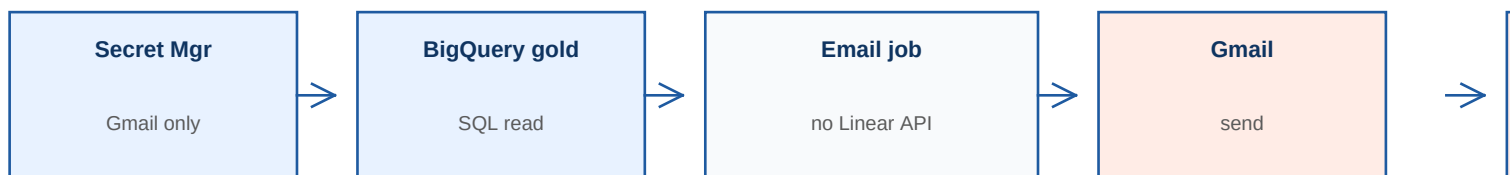
6. TARGET: full pipeline with two crons

When complete, two scheduled jobs run each weekday morning in order:

~6:00 AM - linear-ingest (CRON - planned)



~9:00 AM - daily-sweep-report (CRON - live today, changes in Phase 5)



Main change in Phase 5

The 9 AM email stops calling Linear. It only reads gold views in BigQuery. Linear API is used only by the 6 AM copy job.

7. Side-by-side comparison

Today vs target - ingress and egress per job

Component	Trigger	Ingress (in)	Egress (out)
9 AM email (today)	Cron	IN: Secret Mgr, Linear API	OUT: Gmail email
9 AM email (target)	Cron	IN: Secret Mgr, BQ gold	OUT: Gmail email
Copy job (today)	Manual	IN: Secret Mgr, Linear API	OUT: GCS bronze only
Copy job (target)	Cron ~6 AM	IN: Secret Mgr, Linear API	OUT: GCS + BQ silver
BigQuery gold	Views	IN: silver tables	OUT: read by email job

Network direction summary (external systems)

- Linear API: ingress TO our jobs only (read today; write maybe later)
- Gmail: egress FROM email job only
- Secret Manager: ingress TO jobs (credentials never in code)
- GCS + BigQuery: internal GCP storage between our jobs

8. Where to study in the repo

Component	Trigger	Ingress (in)	Egress (out)
9 AM cron config	Terraform	cloud_run_jobs.tf	scheduler block
Email job code	Python	src/jobs/daily-sweep-report/	linear.py, gmail.py
Copy job code	Python	src/jobs/linear-ingest/	extract.py, gcs.py
Bronze files	GCS	gs://peq-tools-linear-data/bronze/	JSON archive
Architecture	Docs	docs/ARCHITECTURE.md	full design